

SU Jiaji

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EDUCATION

JAN 2024	DOCTOR OF PHILOSOPHY IN STATISTICS Department of Statistics & Data Science <i>National University of Singapore</i> , Singapore THESIS ADVISOR: Dr. YAO Zhigang
JUN 2018	BACHELOR OF NATURAL SCIENCE IN STATISTICS School of Mathematical Science <i>Zhejiang University</i> , Hangzhou, China

PROFESSIONAL EXPERIENCE

MAR 2026 – CURRENT	HONGSHEN YOUNG FACULTY College of Mathematics and Statistics Chongqing University, Chongqing, China	
JUL 2025 – MAR 2026	INDEPENDENT RESEARCHER Chengde, Hebei, China	
OCT 2023 – JUL 2025	RESEARCH FELLOW Department of Statistics & Data Science National University of Singapore, Singapore	Advisor: Dr. YAO Zhigang
AUG 2022 – OCT 2023	STUDENT RESEARCH ASSISTANT Department of Statistics & Data Science National University of Singapore, Singapore	Advisor: Dr. YAO Zhigang

SELECTED ONGOING PROJECTS

SINCE JUN 2025	MANIFOLD FITTING FOR BIOLOGICAL SEQUENCE EMBEDDINGS • Extends our manifold fitting framework to high-dimensional embeddings produced by Transformer-based DNA/RNA language models. • Identifies low-dimensional structures that coherently organise sequences sharing biological functions, providing geometric insight into the “sequence language space.” • Incorporates the fitted manifolds into classification, clustering, and other downstream tasks, enhancing accuracy and interpretability over raw embeddings.
SINCE JAN 2024	LOW-DIMENSION STRUCTURE ANALYSIS BASED ON UK BIOBANK • Applies manifold-fitting methods to large-scale biomedical data, with a focus on the UK Biobank. • Aims to uncover low-dimensional structures within high-dimensional population health data. • Leverages these structures to improve population stratification and refine medical risk assessments.

PREPRINTS & PUBLICATIONS

ACCEPTED:

*Equal Contribution; #Correspondence

- #Yao, Z. & Su, J.. (2026). Manifold fitting: A review of methods and applications. *Harvard Data Science Review*, 8(3).
- *Li, B., *Su, J., *Lin, R., #Yau, S.-T., & #Yao, Z. (2025). Manifold fitting reveals metabolomic heterogeneity and disease associations in UK biobank populations. *Proceedings of the National Academy of Sciences*, 122(22), e2500001122.
- #*Yao, Z., *Su, J., & #*Yau, S.-T. (2024). Manifold fitting with cycleGAN. *Proceedings of the National Academy of Sciences*, 121(5), e2311436121.
- Su, J., #Yao, Z., Li, C., & Zhang, Y. (2023). A statistical approach to estimating adsorption-isotherm parameters in gradient-elution preparative liquid chromatography. *The Annals of Applied Statistics*, 17(4), 3476–3499.

PREPRINTS:

- Su, J. & #Yao, Z. (2025). Principal decomposition with nested submanifolds. *arXiv preprint arXiv:2502.10010*. (*Biometrika Major Revision*).
- #Yao, Z., Su, J., Li, B., & Yau, S.-T. (2023). Manifold fitting. *arXiv preprint arXiv:2304.07680*. (*JMLR Under Review*).

GRATNS & AWARDS

APR 2026 | CHONGQING UNIVERSITY STARTUP GRANT

TALKS & PRESENTATIONS

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|---------------------------------|---|
| JUN 2026
Lanzhou | CHINESE ASSOCIATION FOR APPLIED STATISTICS
<i>Transition of the Professional Committee on Multivariate Analysis Application 2026 Annual</i> |
| OCT 2025
Chongqing | TALK: ‘MANIFOLD FITTING AND PRINCIPAL NESTED SUBMANIFOLDS’
<i>College of Mathematics and Statistics, Chongqing University</i> |
| MAY 2025
Beijing | YMSC STATISTICAL SEMINAR
<i>Yau Mathematical Sciences Center, Tsinghua University</i> |
| JAN 2025
Shanghai
& Sanya | THE SECOND SYMPOSIUM OF GEOMETRY AND STATISTICS IN CHINA
<i>Shanghai Institute for Mathematics and Interdisciplinary Sciences
& The Tsinghua Sanya International Mathematics Forum</i> |
| OCT 2024
Singapore | INTERACTIONS OF STATISTICS AND GEOMETRY (ISAG) II
<i>Institute for Mathematical Sciences, National University of Singapore</i> |
| JUN 2024
Shanghai | SUMMER SEMINAR SERIES: STATISTICS + GEOMETRY + X
<i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i> |
| JUL 2023
Beijing | TALK: ‘ESTIMATING ADSORPTION-ISOTHERM PARAMETERS’
<i>School of Mathematics and Statistics, Beijing Institute of Technology</i> |

VISITING

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|------------------------|---|
| MAY 2026
– JUN 2026 | VISITING RESEARCHER
Shanghai Institute for Mathematics and Interdisciplinary Sciences, China |
| DEC 2024
– JAN 2025 | VISITING RESEARCHER
Shanghai Institute for Mathematics and Interdisciplinary Sciences, China |
| JUN 2024
– JUL 2024 | VISITING RESEARCHER
Shanghai Institute for Mathematics and Interdisciplinary Sciences, China |

TEACHING

Teaching assistant at National University of Singapore:

Semester	Courses
2021-22 Sem 2	ST2334 Probability and Statistics
	DSA1101 Introduction to Data Science
	ST2137 Statistical Computing and Programming
2021-22 Sem 1	ST5213 Advanced Categorical Data Analysis
	DSA1101 Introduction to Data Science
2020-21 Sem 2	ST2334 Probability and Statistics
	ST3232 Design and Analysis of Experiments
2020-21 Sem 1	ST2334 Probability and Statistics
	ST5225 Statistical Analysis of Networks
2019-20 Sem 2	ST2131 Probability
	ST3239 Survey Methodology
2019-20 Sem 1	ST4231 Computer Intensive Statistical Methods
2018-19 Sem 2	ST2131 Probability